

PERFORMANCE TESTING

Date	11 October 2026
Team ID	SWTID-2026-8809
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Team Members	N.Pooja, T.Sangeetha, K.Divya, Veronica, N.Monika
Maximum Marks	5 Marks

Performance Testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust / Postman / FastAPI TestClient
Type of Testing	Load Testing & API Latency Performance Benchmarking
Target Module / API	/qa, /explain, /quiz, /summarize, /learn/recommendations
Test Environment	Local Server (Uvicorn ASGI, macOS M1 / Python 3.10)
Test Date	11 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Concurrent Q&A requests via Gemini 1.5 Pro API (GET /qa)	10	60	Response time < 2.0s, 0% error rate
2	Local concept explanation generation using LaMini-Flan-T5 (POST /explain)	5	60	Response time < 1.5s, CPU < 80%
3	Structured Quiz Generation and JSON parsing (POST /quiz)	10	60	Valid JSON output, Response < 3.0s
4	Text Summarization & Adaptive Learning Path (POST /summarize, GET /learn/recommendations)	15	120	Successful response streaming, zero dropouts

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.24 seconds	PASS	Optimal
2	Response Time (Max)	< 5 seconds	3.10 seconds	PASS	Within bounds
3	Throughput (Req/sec)	> 10 req/sec	14.2 req/sec	PASS	High efficiency
4	Error Rate	< 1%	0.00%	PASS	Zero failures
5	CPU Utilization	< 80%	42%	PASS	Well managed
6	Memory Utilization	< 80%	58%	PASS	Stable load

Step 4: Observations & Analysis

Key Findings:

The performance test results indicate that the EduGenie core application performs efficiently under multi-user loads. Cloud-based API endpoints leveraging Google Gemini 1.5 Pro achieved low latency with average response times well within the 2-second benchmark. Local CPU execution of the lightweight LaMini-Flan-T5 model remained stable under 60% memory footprint.

Bottlenecks Identified:

Slight response latency variations were observed during complex JSON block parsing in high-frequency quiz generation requests, primarily due to cloud API network round-trip times during peak execution.

Optimization Steps Taken:

Implemented asynchronous endpoint handlers in FastAPI using Uvicorn ASGI workers. Added efficient regex cleaning ('clean_json_block') and lightweight prompt structures to minimize token overhead and cloud latency.

Step 5: Screenshots / Evidence

[Evidence 1: Terminal Server Logs & Execution Verification]

```
INFO:      Started server process [17751]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
INFO:      Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:      127.0.0.1:54321 - "GET /qa?question=Which+is+the+largest+ocean? HTTP/1.1" 200 OK - 1.12s
INFO:      127.0.0.1:54322 - "POST /quiz HTTP/1.1" 200 OK - 2.04s
```

[Evidence 2: Functional API Endpoint Test Summary]

```
===== Performance Summary =====  
Target: http://127.0.0.1:8000  
Endpoints Tested: /qa, /explain, /summarize, /quiz, /learn/recommendations  
Total Requests: 500 | Passed: 500 | Failed: 0 | Error Rate: 0.00%  
Average Response Time: 1.24s | Max Response Time: 3.10s  
Status: SYSTEM READY AND PERFORMANCE VERIFIED  
=====
```